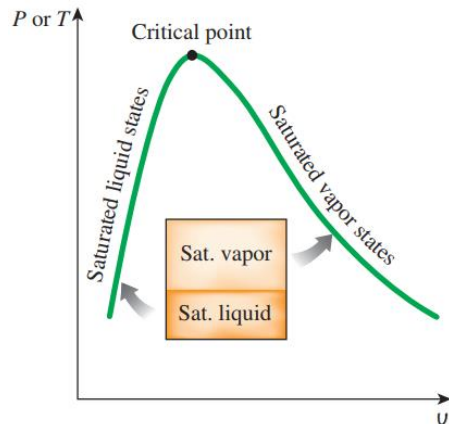


CHAPTER 2: VAPOR AND COMBINED POWER CYCLES

- We looked at Gas power cycles (Otto, Diesel ,Stirling and Brayton) which were heat engines for mostly mobile applications.

Vapor power cycles are used in:

- Steam engines (mobile)
- Power plants (stationary)
- the working fluid is alternately vaporized and condensed.
- Steam is the most common working fluid used in vapor power cycles because of its many desirable characteristics, such as **low cost, availability, and high enthalpy of vaporization.**



$$x = \frac{m_{\text{vapor}}}{m_{\text{total}}}$$

$$m_{\text{total}} = m_{\text{liquid}} + m_{\text{vapor}} = m_f + m_g$$

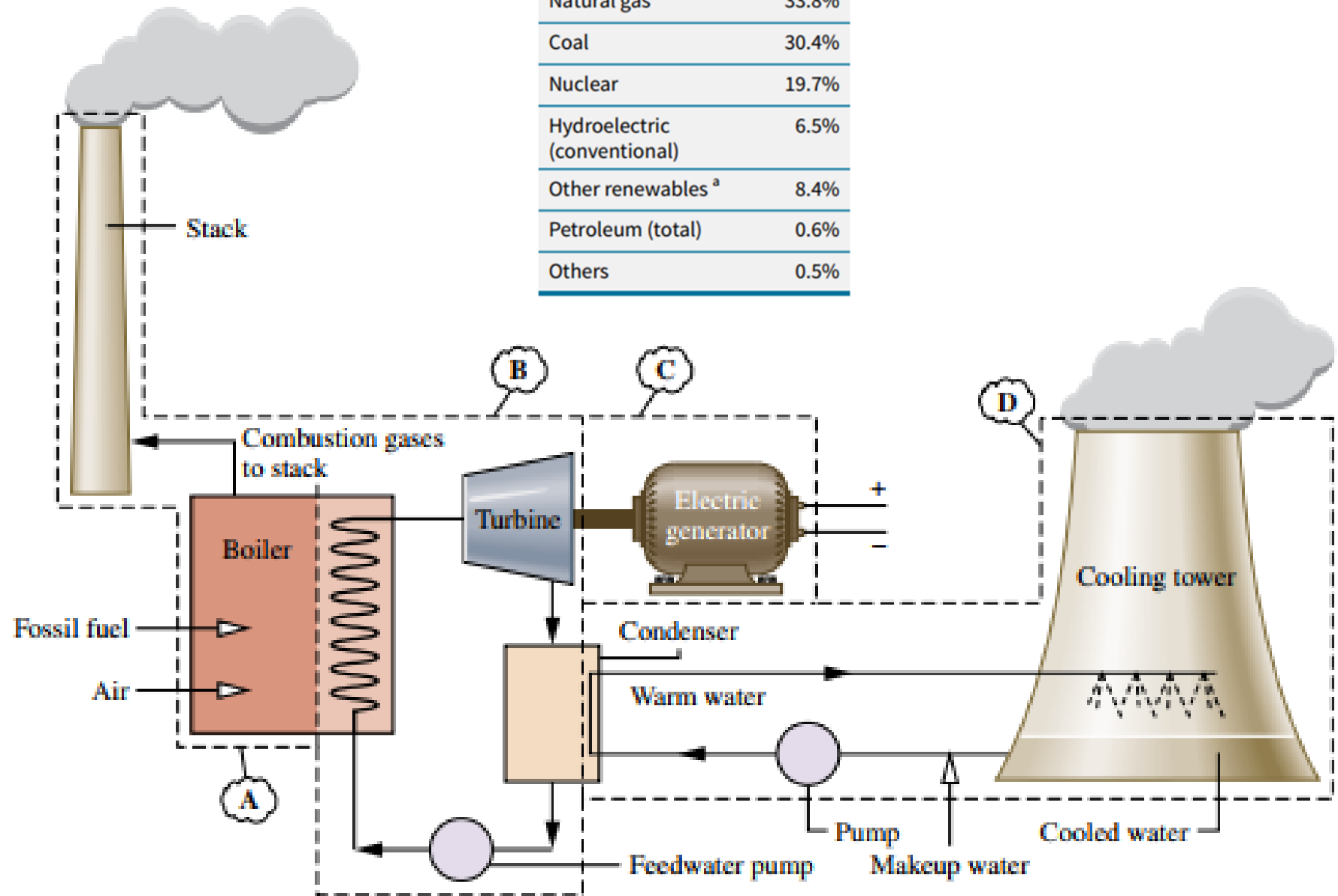
$$h_{\text{avg}} = h_f + xh_{fg} \quad (\text{kJ/kg})$$

Enthalpy of vaporization:

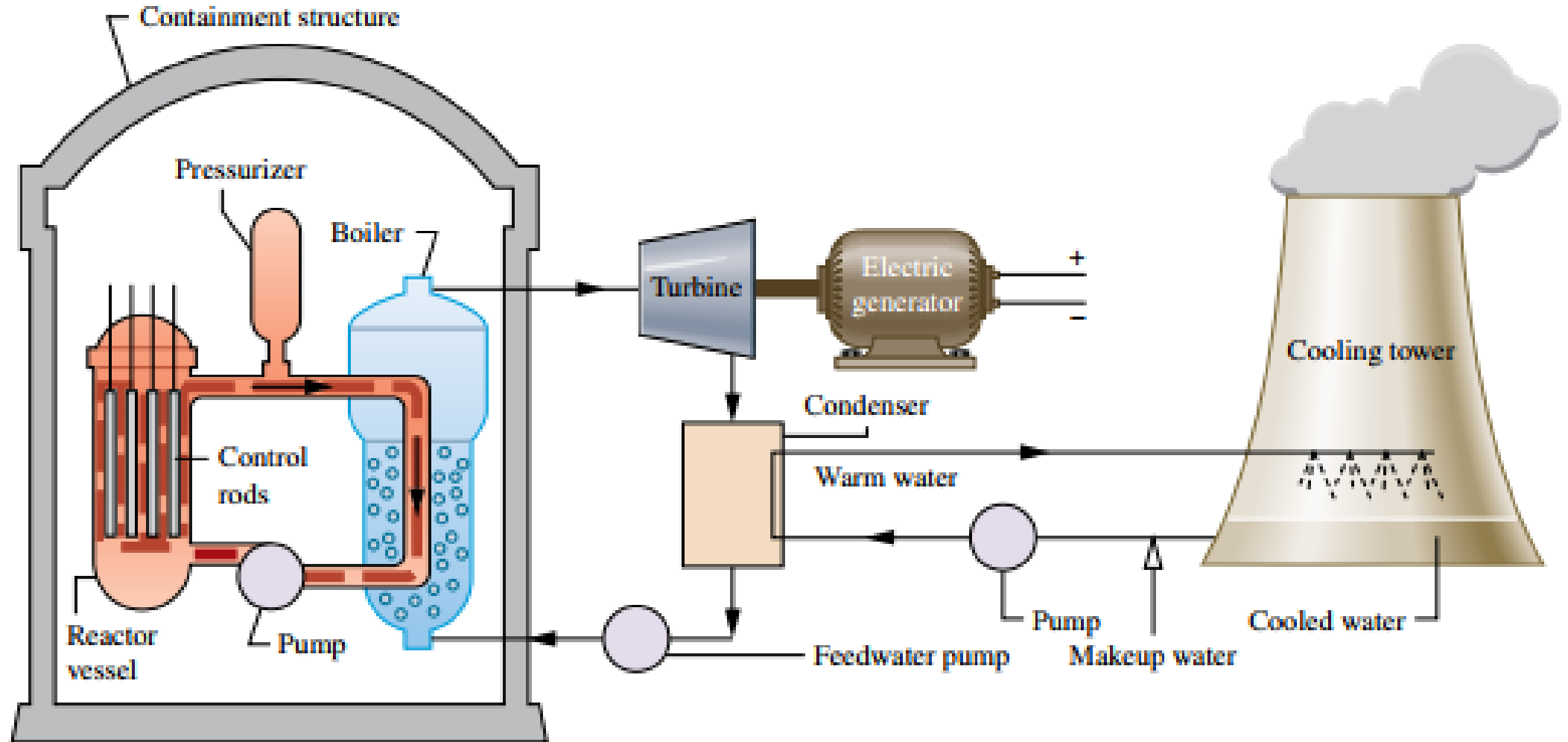
Steam has a high enthalpy of vaporization, meaning it can store a large amount of energy when it changes from a liquid to a gas

Steam power plants are commonly referred to as *coal plants*, *nuclear plants*, or *natural gas plants*, depending on the type of fuel used to supply heat to the steam.

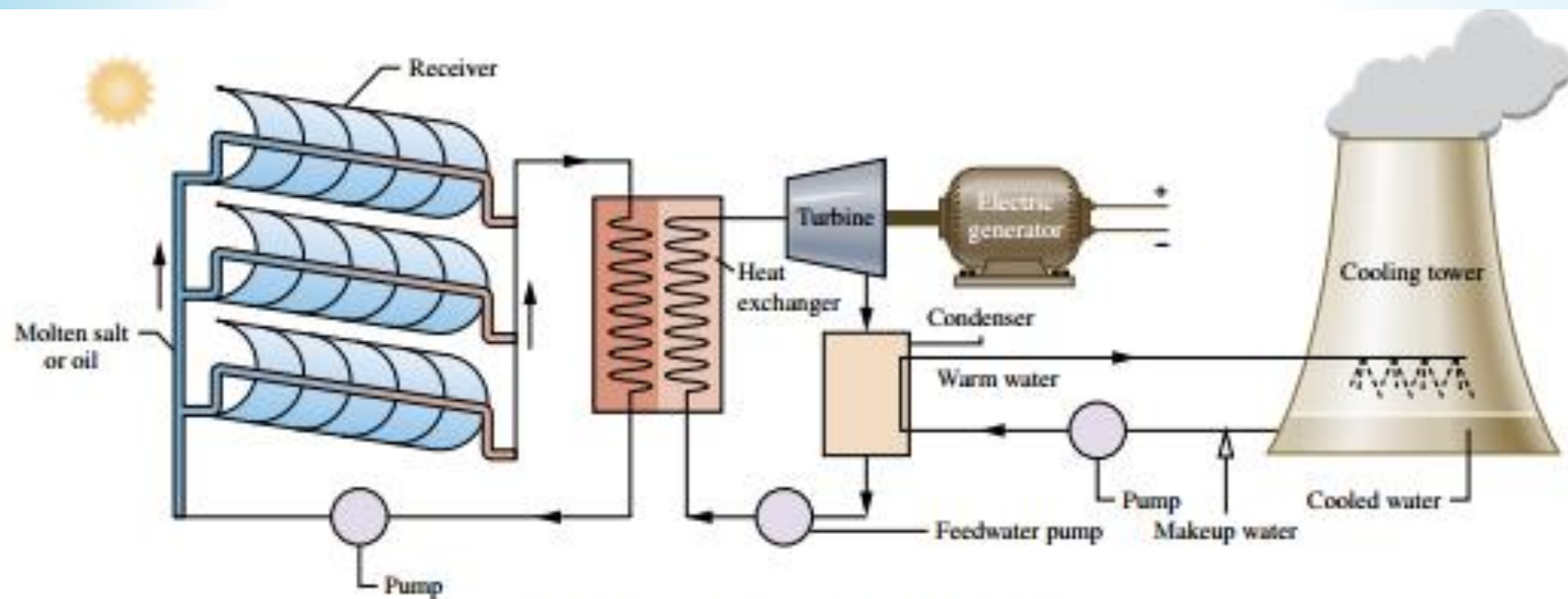
Natural gas	33.8%
Coal	30.4%
Nuclear	19.7%
Hydroelectric (conventional)	6.5%
Other renewables ^a	8.4%
Petroleum (total)	0.6%
Others	0.5%



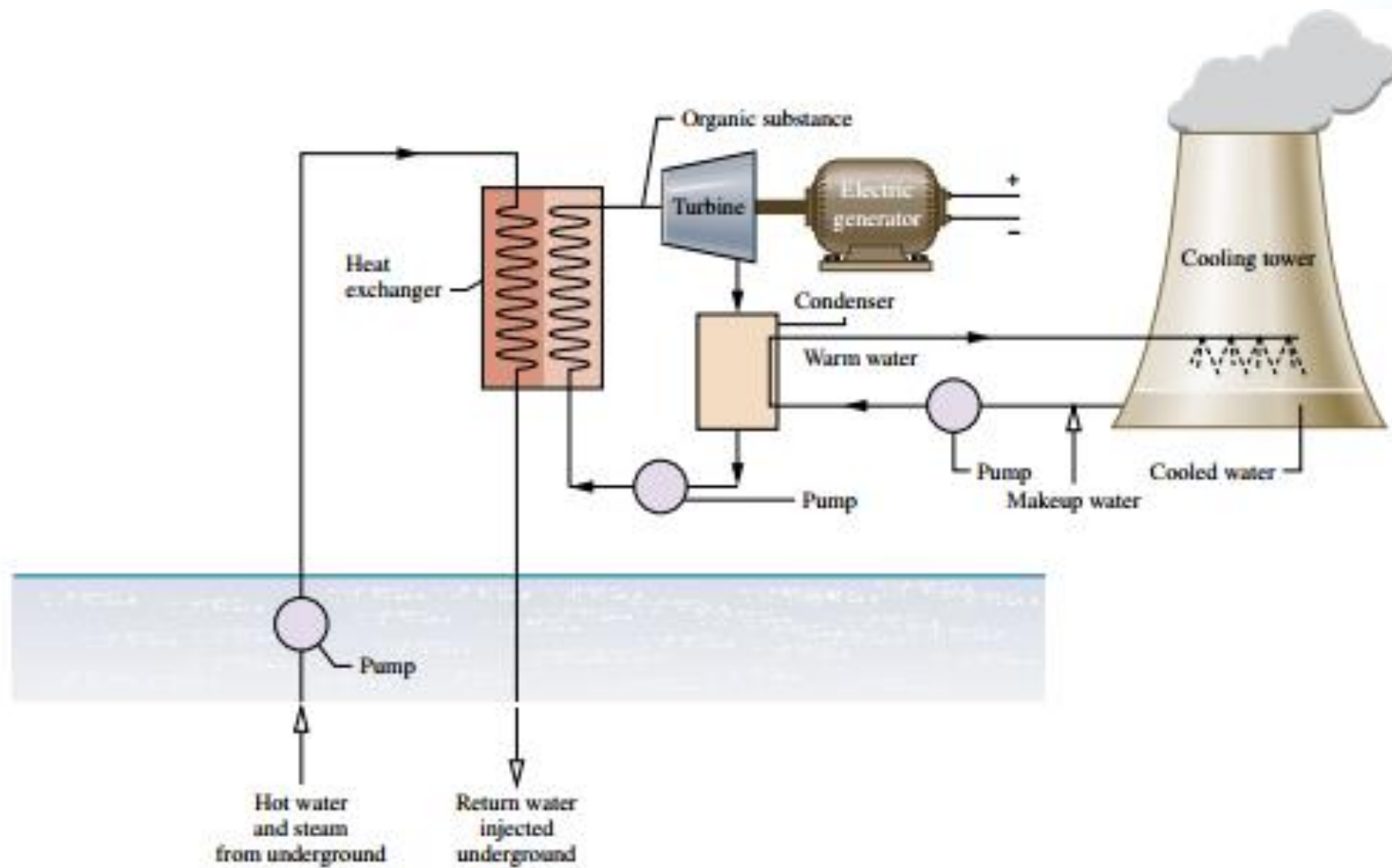
(a) Fossil-fueled vapor power plant.



(b) Pressurized-water reactor nuclear vapor power plant.



(c) Concentrating solar thermal vapor power plant.



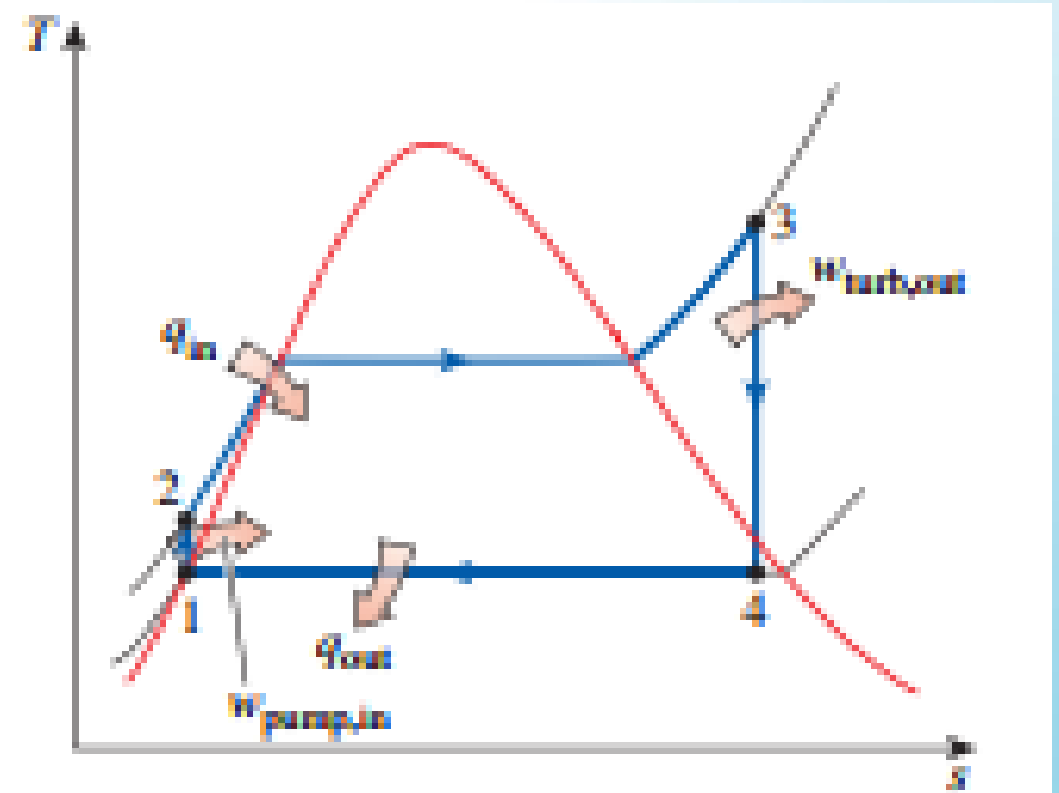
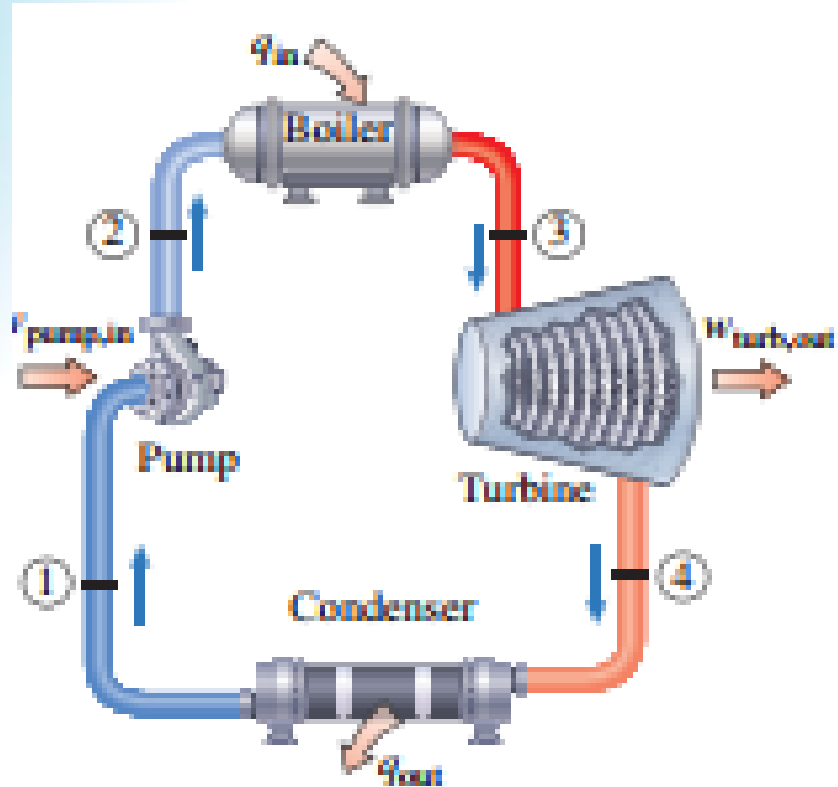
(d) Geothermal vapor power plant.

THE RANKINE CYCLE

The Rankine cycle is the basic building block of vapor power plants.

The ideal Rankine cycle does not involve any internal irreversibilities and consists of the following four processes:

- 1-2 Isentropic compression in a pump
- 2-3 Constant-pressure heat addition in a boiler
- 3-4 Isentropic expansion in a turbine
- 4-1 Constant-pressure heat rejection in a condenser



The simple ideal Rankine cycle.

ENERGY ANALYSIS OF THE IDEAL RANKINE CYCLE

$$W_{net} = W_{out} - W_{in} = q_{in} - q_{out}$$

Applying the first law for steady-flow device to each component:

$$(q_{in} - q_{out}) - (w_{out} - w_{in}) = h_e - h_i$$

Pump: 1 → 2 (q=0)

$$w_{pump,in} = h_2 - h_1$$

$$w_{pump,in} = v(p_2 - p_1)$$

$$h_1 = h_f @ p_1 \quad \text{and } v = v_1 = v_f @ p_1$$

Boiler: 2 → 3 (w=0)

$$q_{in} = h_3 - h_2$$

Turbine: 3 → 4 (q=0)

$$w_{turb,out} = h_3 - h_4$$

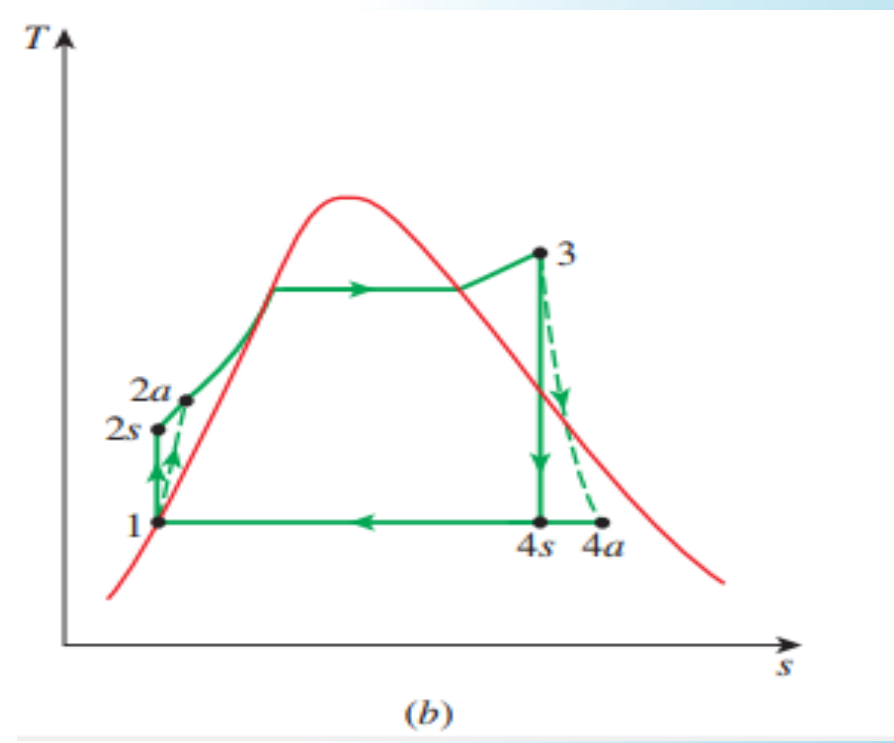
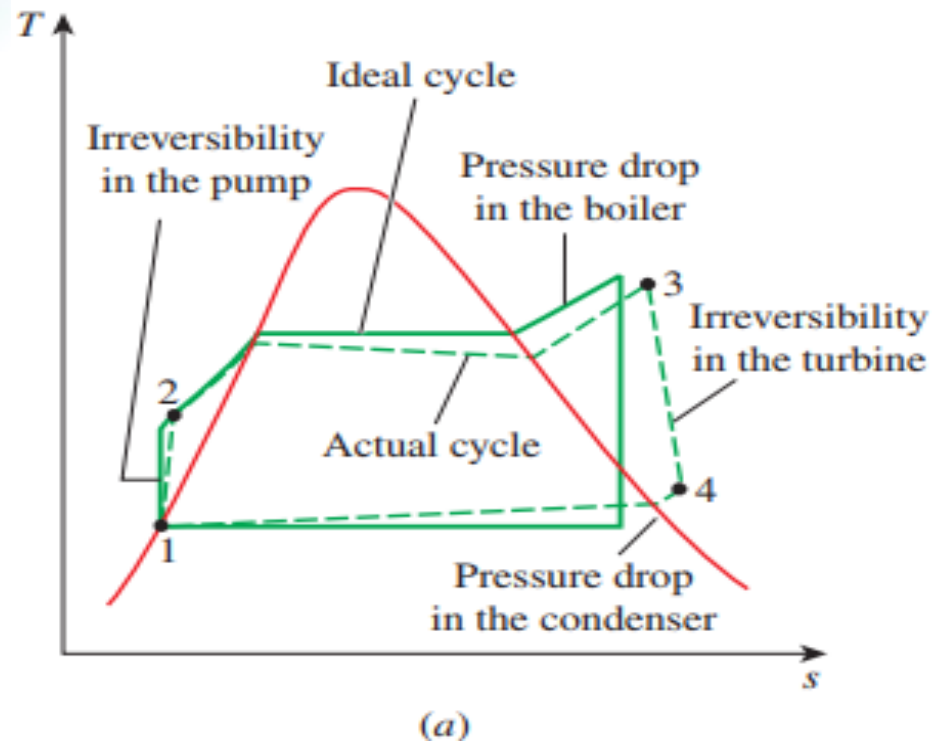
Condenser: 4 → 1 (w=0)

$$q_{out} = h_4 - h_1$$

Thermal Efficiency: $\eta_{th} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$

ACTUAL RANKINE CYCLE (DEVIATION OF THE ACTUAL VAPOR CYCLES FROM IDEALIZED ONES)

Real steam power plants suffer from fluid friction (pressure drop in heat exchangers) and no-isentropic expansion and compression.



- Fluid friction causes pressure drops in the boiler, the condenser, and the piping between various components.
As a result, steam leaves the boiler at a somewhat lower pressure.
- Also, the pressure at the turbine inlet is somewhat lower than that at the boiler exit due to the pressure drop in the connecting pipes.
- The pressure drop in the condenser is usually very small.
- To compensate for these pressure drops, the water must be pumped to a sufficiently higher pressure than the ideal cycle calls for.
- This requires a larger pump and larger work input to the pump.
- The deviation of actual pumps and turbines from the isentropic ones can be accounted for by utilizing isentropic efficiencies, defined as:

$$\eta_p = \frac{w_s}{w_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

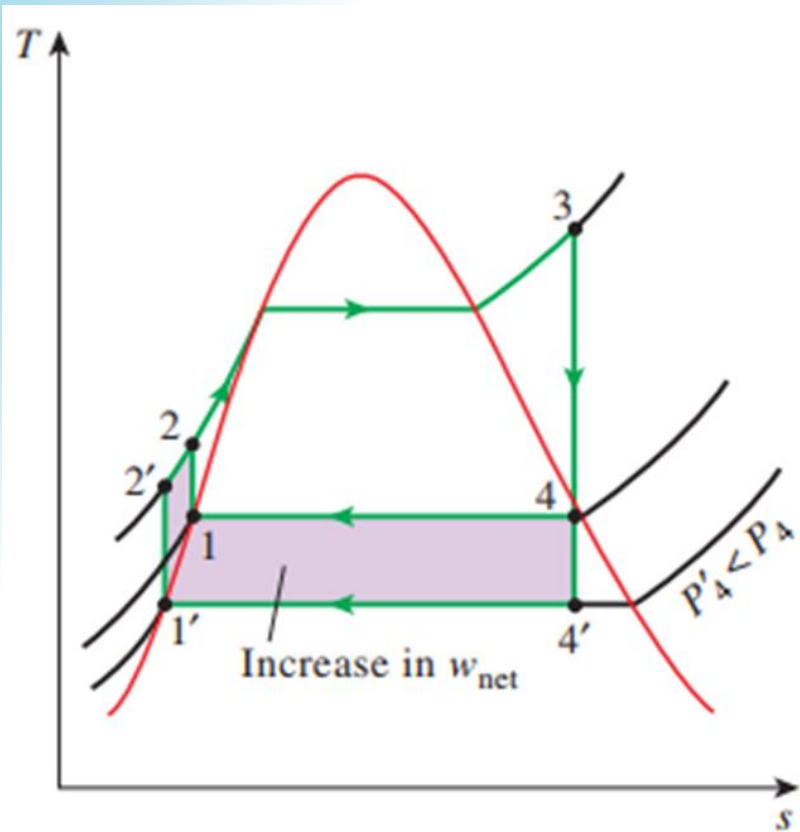
and

$$\eta_T = \frac{w_a}{w_s} = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

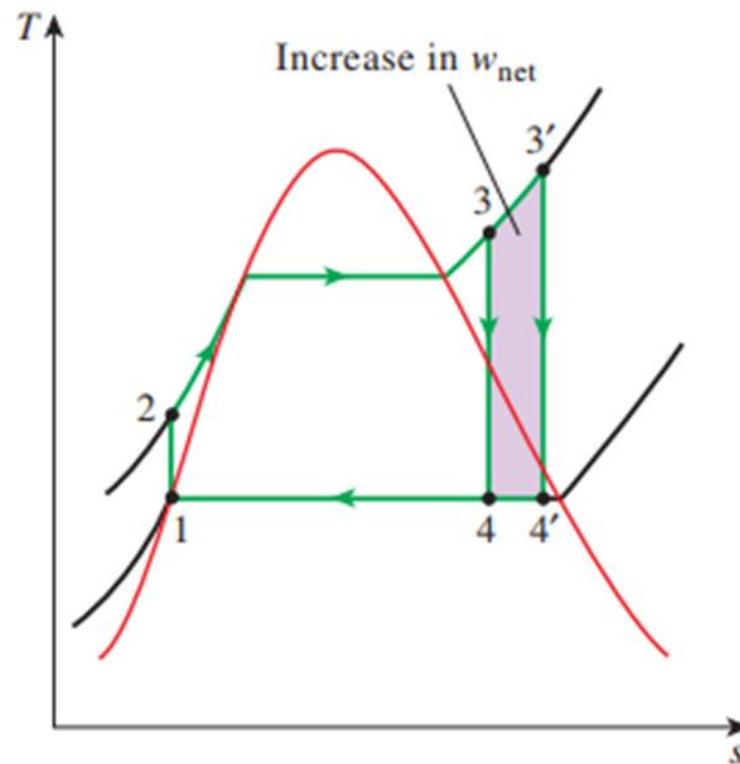
HOW CAN WE INCREASE THE EFFICIENCY OF THE RANKINE CYCLE?

We can increase w_{net} and η_{th} by:

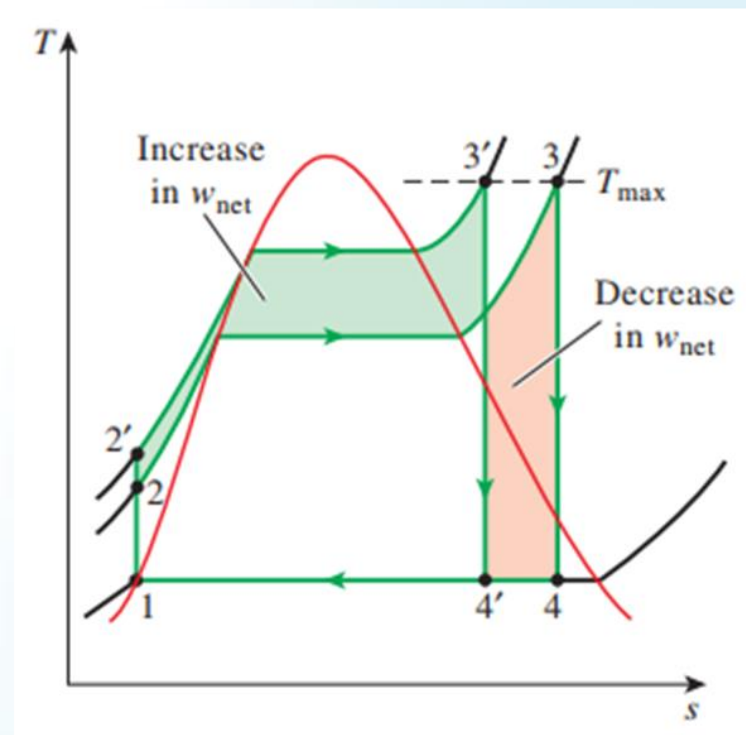
- Reducing the condenser pressure.
 - A lower condenser pressure reduces the temperature at which heat is rejected.
 - This increases the enthalpy drop across the turbine, allowing it to extract more work from the steam.
- Superheat the fluid to higher temperature.
 - Superheating raises the temperature at which heat is added, increasing the average temperature of heat input.
- Raise the boiler pressure,
 - Raising boiler pressure elevates the temperature at which heat is added to the working fluid.
 - This increases the average



The effect of lowering the condenser pressure on the ideal Rankine cycle.



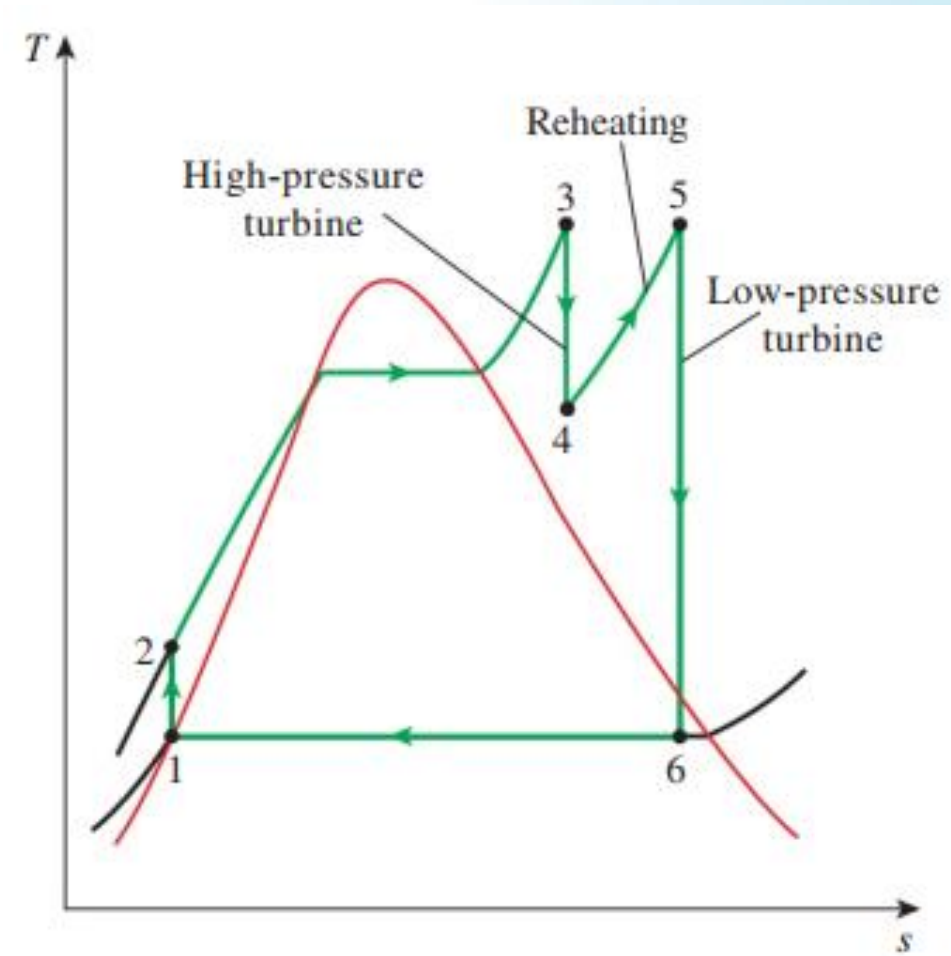
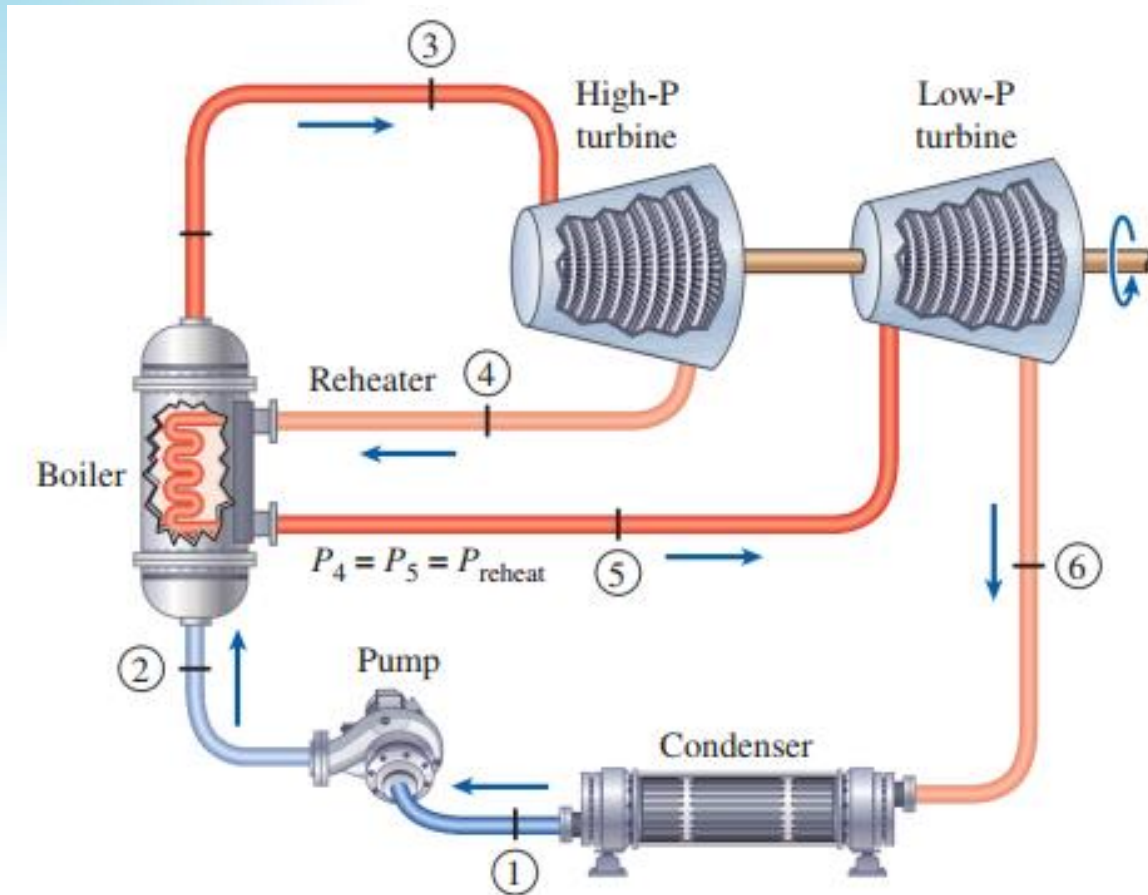
The effect of superheating the steam to higher temperatures on the ideal Rankine cycle



The effect of superheating the steam to higher temperatures on the ideal Rankine cycle

RANKINE CYCLE WITH REHEAT

- Increases the area under the curve, and thus w_{net} and η_{th} , and prevents the problems associated with passing wet steam through the turbine.



The ideal reheat Rankine cycle.

In actual power plant, the efficiency increase is due to:

- 40% of the improvement is from increased w_{net} area.
- 60% from increased turbine efficiency due to reduced moisture loss.
- Reheat pressure is usually 0
- 0.2 to 0.3 time the initial pressure
- Due economics, only one reheat stage is typically employed.

REGENERATIVE RANKINE CYCLE

- A regenerator or a feed water heater takes some steam from the turbine and uses it to heat water after the pump(state 2).
- This increases thermal efficiency as we are adding heat at a higher temperature
- Recall the Carnot Efficiency

$$\eta_{th} = 1 - \frac{T_l}{T_H}$$

Raise the temperature at which heat is added and η_{th} goes up.

There are two types of feedwater heater (regenerator):

- 1) Open Feedwater Heater (FWH)
- 2) Closed feedwater Heater (FWH)

The heaters also raise water temperature after the pump in order to avoid thermal shock to the boiler metal.

OPEN FEEDWATER HEATER

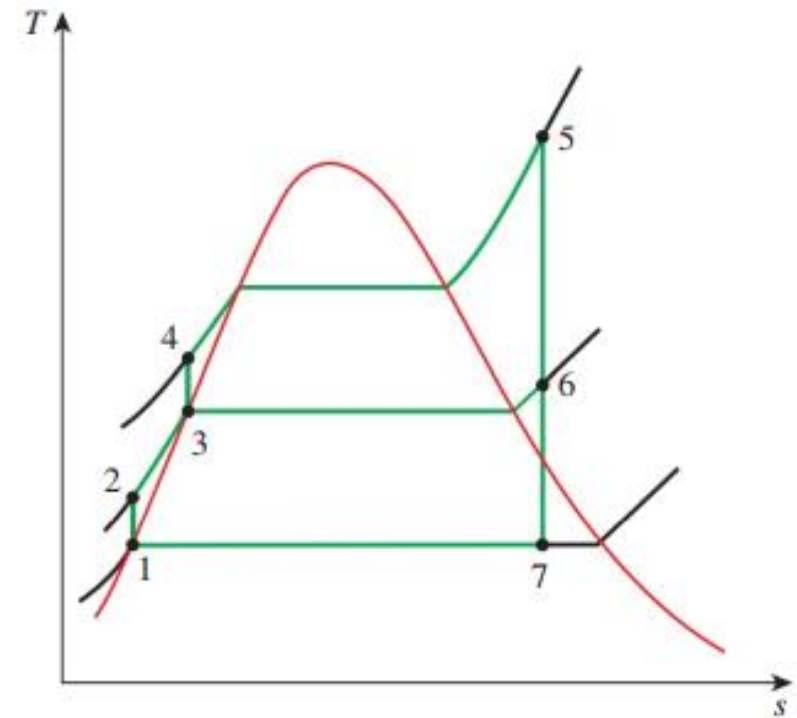
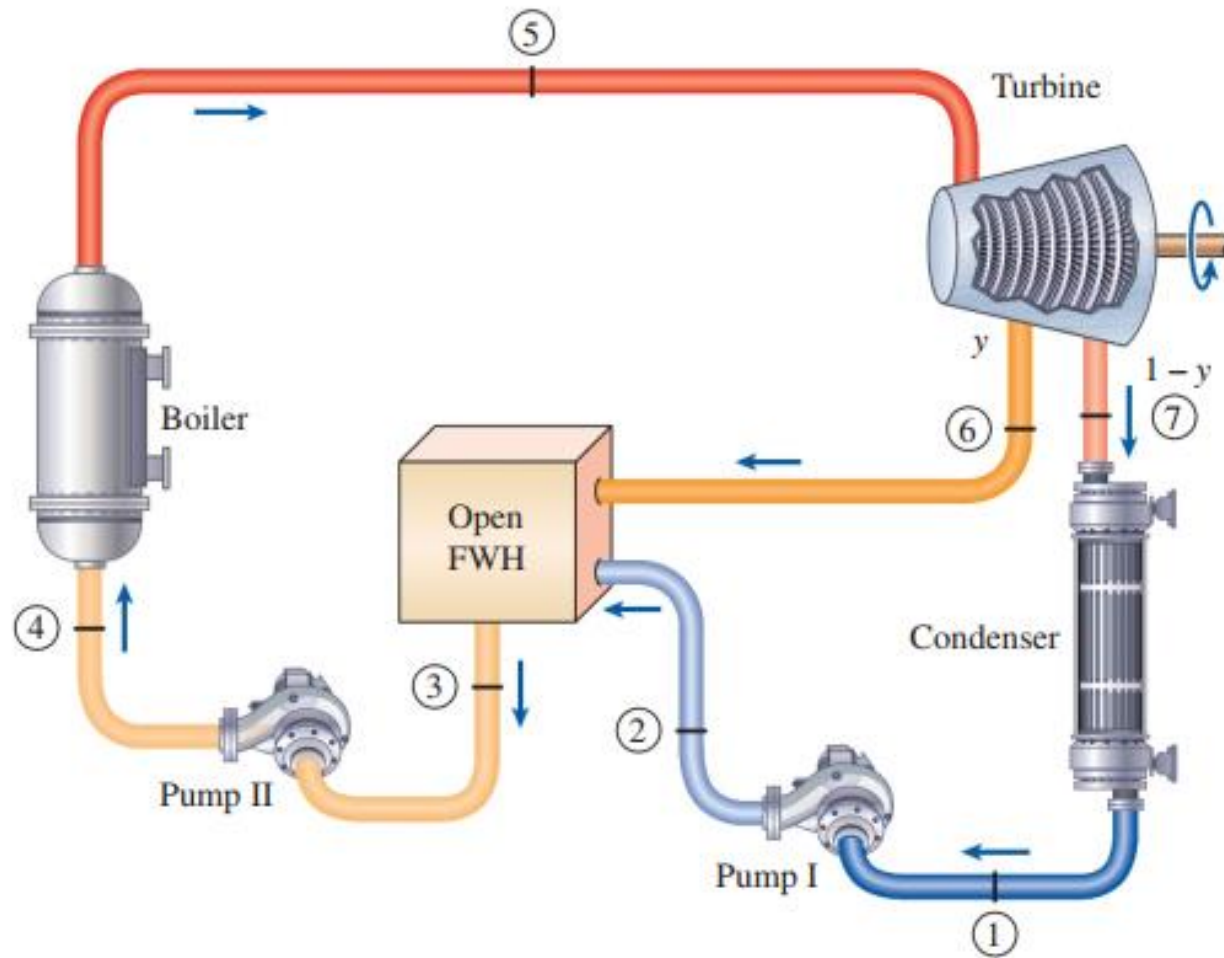
- An open (or direct-contact) feedwater heater is a mixing chamber designed to improve the efficiency of the Rankine cycle.

Process

- Steam is extracted from the turbine at an intermediate pressure.
- This extracted steam is routed to the feedwater heater, where it mixes directly with the feedwater (condensate from the condenser) that has been pumped to the same pressure.
- Ideally, the mixture leaves the heater as a saturated liquid at the heater pressure.

SINGLE STAGE REGENERATIVE CYCLE

- Steam enters the turbine at boiler pressure.
- It expands isentropically; a portion is extracted for the feedwater heater at an intermediate pressure.
- The remaining steam continues to expand in the turbine to the condenser pressure.
- The condensed liquid is pumped to the feedwater heater pressure, mixes with the extracted steam, and the cycle repeats.
- This "regeneration" process pre-heats the feedwater using internal steam, reducing irreversibilities and increasing the cycle's thermal efficiency.



The ideal regenerative Rankine cycle with an open feedwater heater.

Pump I takes the water up to a pressure such that it equals the pressure of the bleed steam.

Apply the first law to the open FWH:

$$\sum \dot{m}_e h_e = \sum \dot{m}_i h_i$$

$$\dot{m}_3 h_3 = \dot{m}_2 h_2 + \dot{m}_6 h_6 \quad (I)$$

$$\text{Now } \dot{m}_3 = \dot{m}_4 = \dot{m}_5$$

$$\dot{m}_7 = \dot{m}_1 = \dot{m}_2 = \dot{m}_5 - \dot{m}_6$$

We can rewrite (I) as :

$$\dot{m}_5 h_3 = (\dot{m}_5 - \dot{m}_6) h_2 + \dot{m}_6 h_6$$

Divide through by $\dot{m}_5$;

$$h_3 = (1 - y) h_2 + y h_6$$

Other equations

$$q_{in} = h_5 - h_4$$

$$q_{out} = (1 - y)(h_7 - h_1)$$

$$W_{turb,out} = (h_5 - h_6) + (1 - y)(h_6 - h_7)$$

$$W_{pump,in} = (1 - y) W_{pump I,in} + W_{pump II,in}$$

$$y = \frac{\dot{m}_6}{\dot{m}_5}$$

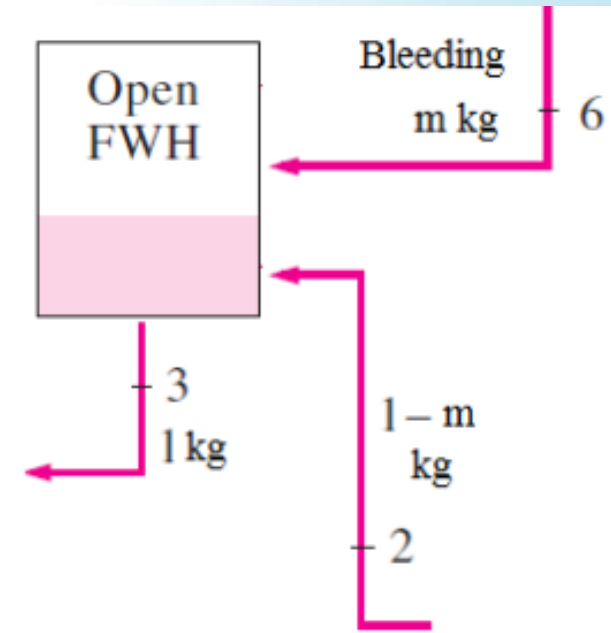
$$W_{pump I,in} = v_1(p_2 - p_1) = h_2 - h_1$$

$$W_{pump II,in} = v_3(p_4 - p_3) = h_4 - h_3$$

Energy Balance

Assume 1 kg of working fluid circulated

Energy in = Energy out



$$m = \frac{\text{mass of steam bled}}{\text{mass of steam circulated}}$$

Energy entering the regenerator = Energy leaving the regenerator

$$mh_6 + (1 - m)h_2 = 1 \times h_3$$

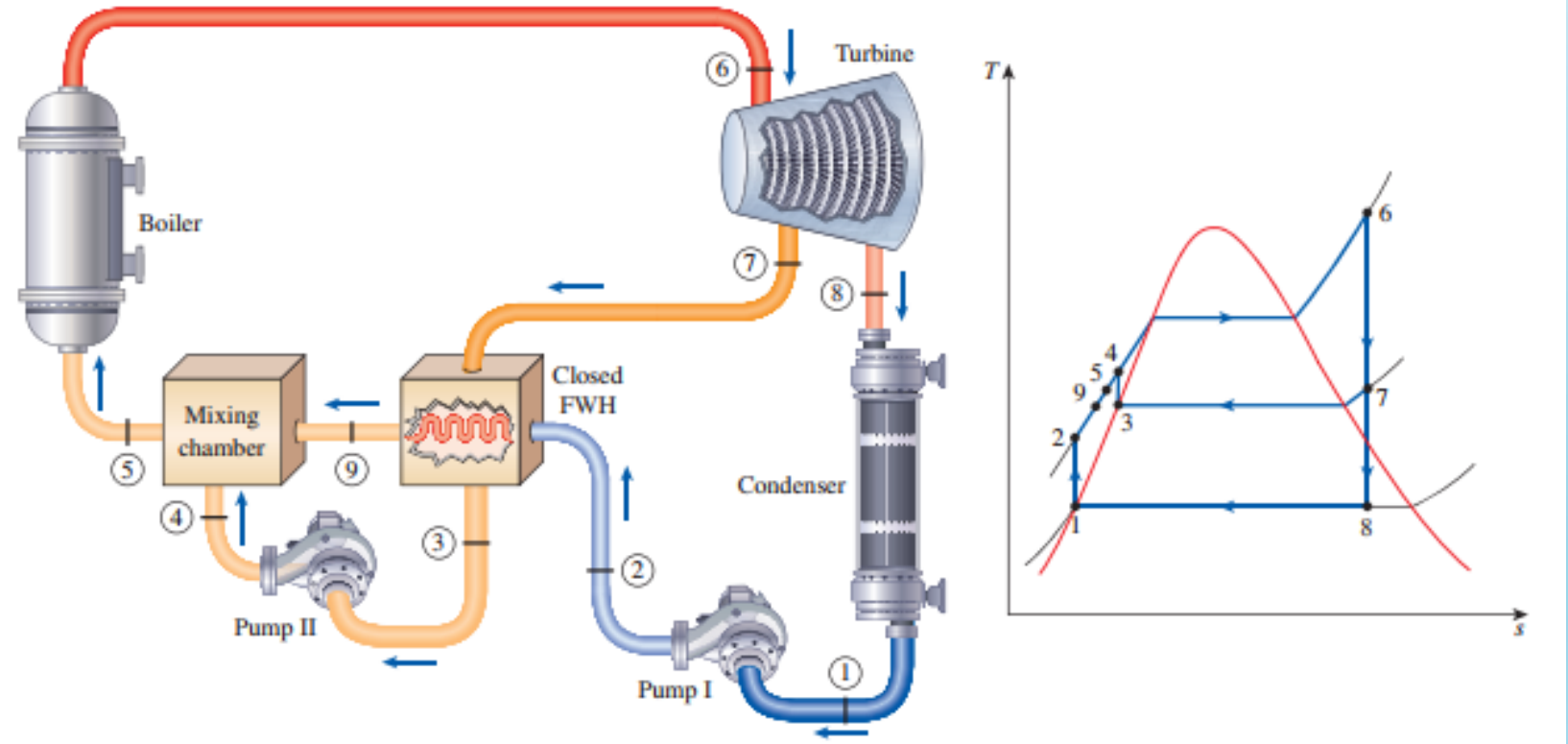
$$mh_6 + h_2 - mh_2 = h_3$$

$$m(h_6 - h_2) = h_3 - h_2$$

$$m = \frac{h_3 - h_2}{h_6 - h_2} \quad \text{mass of steam bled}$$

CLOSED FEEDWATER HEATER

- Basically just a shell and tube heat exchanger.
- In this type, the heat is transferred from the extracted (bled) steam to the feed water without any mixing taking place.
- Does not deaerate.



The ideal regenerative Rankine cycle with a closed feedwater heater.

$$q_{in} = h_6 - h_5$$

$$q_{out} = (1-y)(h_8 - h_1)$$

$$w_{turb,out} = (h_6 - h_7) + (1-y)(h_7 - h_8)$$

$$w_{pump,in} = (1-y) v_1(p_2 - p_1) + y v_3(p_4 - p_3)$$

$$y = \frac{\dot{m}_7}{\dot{m}_6}$$

Apply the first law to the Closed FWH:

$$\sum \dot{m}_e h_e = \sum \dot{m}_i h_i$$

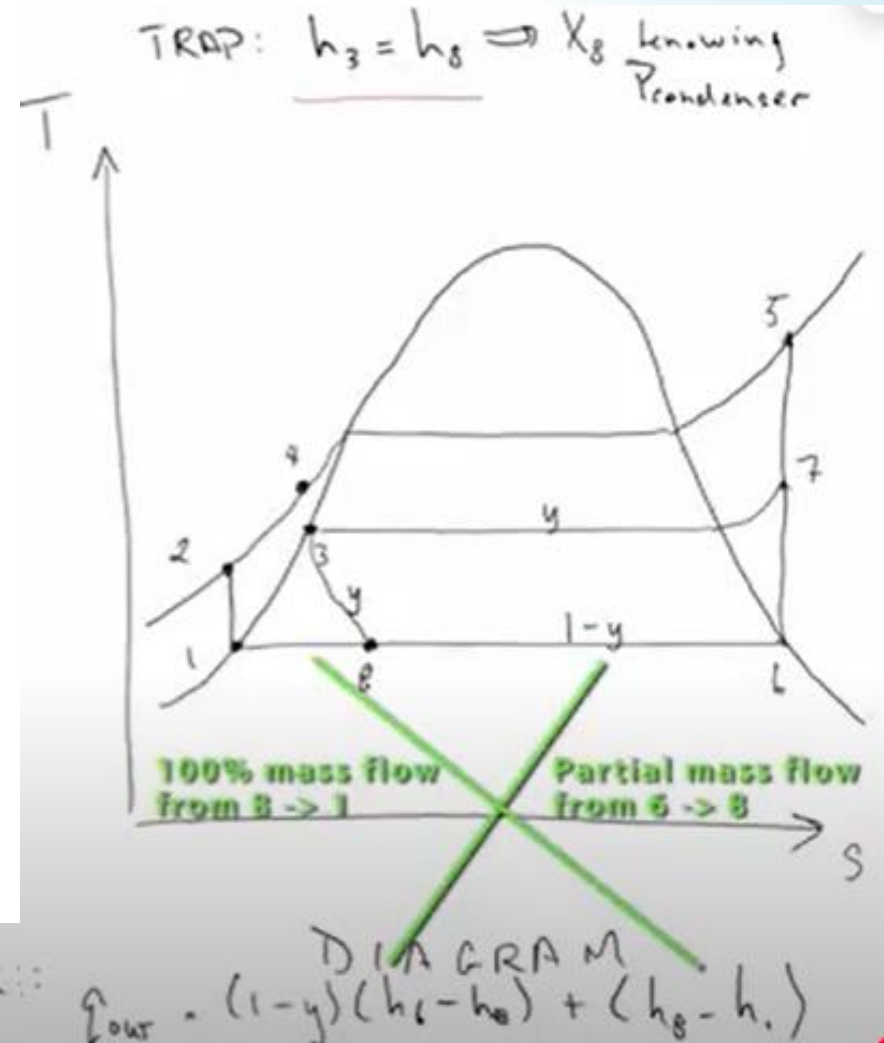
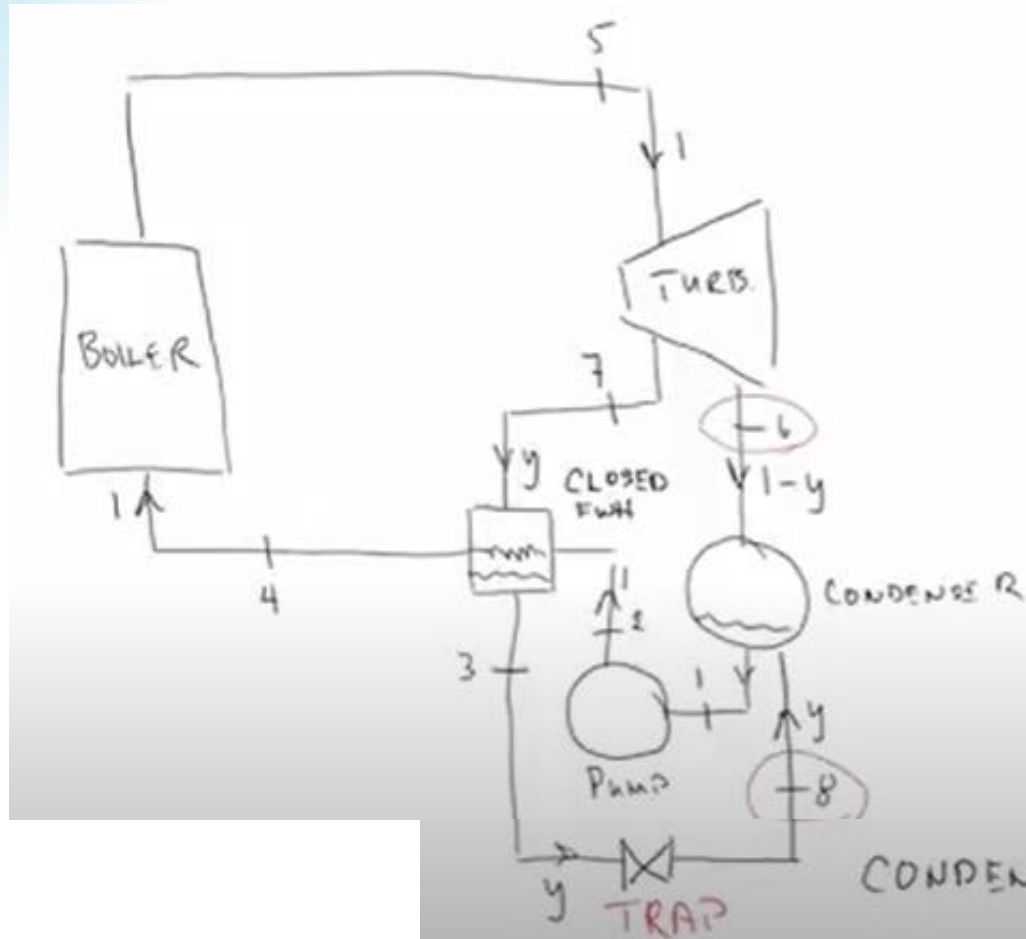
$$(1-y)h_9 + yh_3 = yh_7 + (1-y)h_2$$

Applying first law to the mixing chamber

$$\sum \dot{m}_e h_e = \sum \dot{m}_i h_i$$

$$(1)h_5 = yh_1 + (1-y)h_9$$

- We some times use a trap to throttle a liquid to a lower pressure.
- This is a constant enthalpy process.

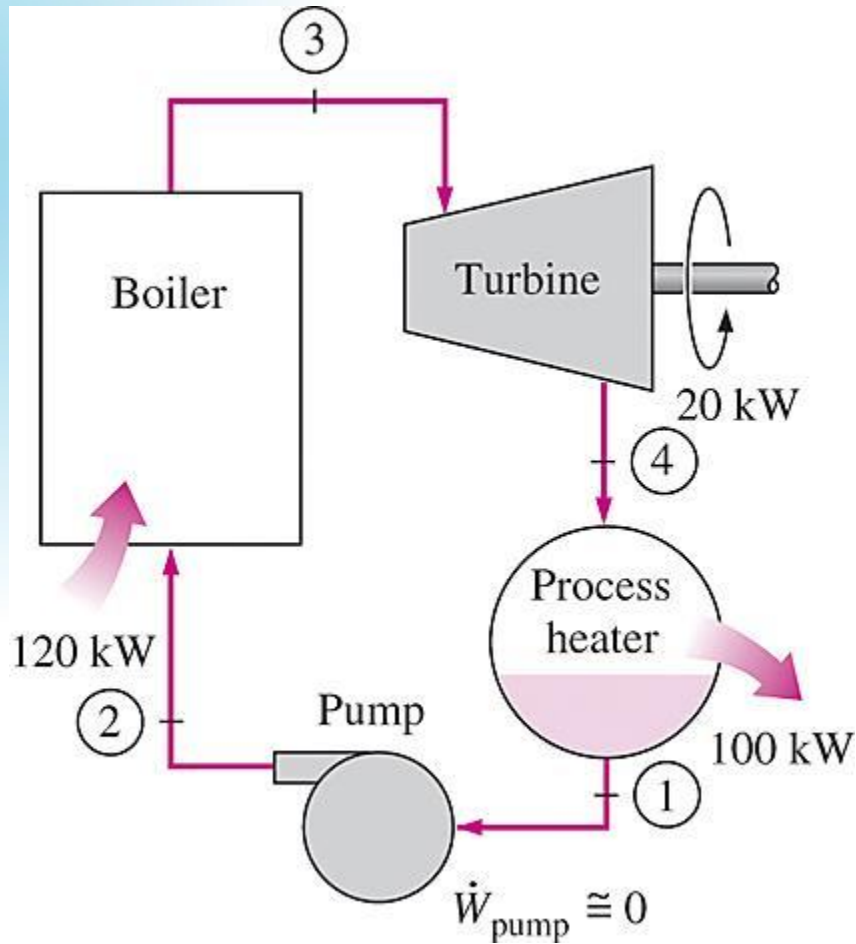


COGENERATION

- Many industries require energy input in the form of heat, called process heat. Process heat in these industries is usually supplied by steam at 5 to 7 atm and 150 to 200°C.
- Energy is usually transferred to the steam by burning coal, oil, natural gas, or another fuel in a furnace.
- Industries that use large amounts of process heat also consume a large amount of electric power.
It makes sense to use the already existing work potential to produce power instead of letting it go to waste.
- The result is a plant that produces electricity while meeting the process-heat requirements of certain industrial processes (cogeneration plant)
- **Cogeneration:** The production of more than one useful form of energy (such as process heat and electric power) from the same energy source.

Utilization

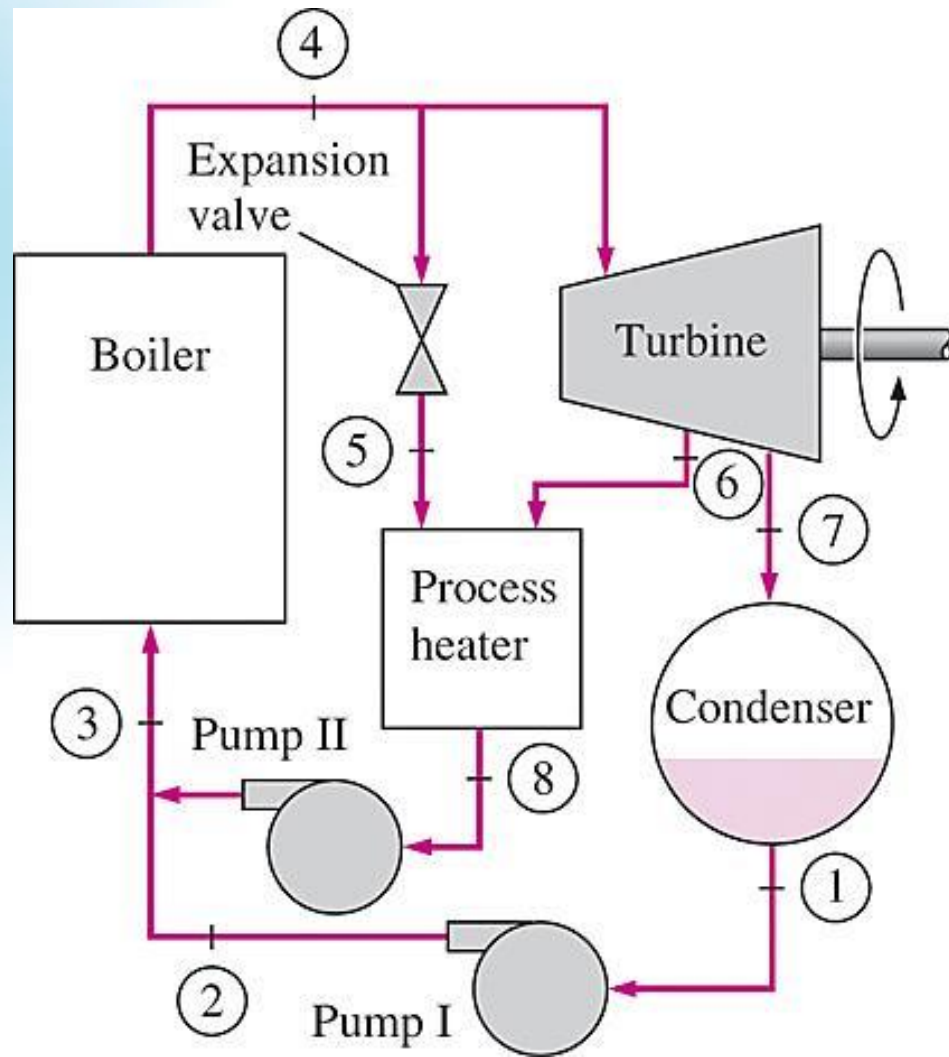
- factor
$$\epsilon_u = \frac{\text{Net work output} + \text{Process heat delivered}}{\text{Total heat input}} = \frac{\dot{W}_{\text{net}} + \dot{Q}_p}{\dot{Q}_{\text{in}}}$$



$$\epsilon_u = 1 - \frac{\dot{Q}_{\text{out}}}{\dot{Q}_{\text{in}}}$$

- The utilization factor of the ideal steam-turbine cogeneration plant is 100%.
- Actual cogeneration plants have utilization factors as high as 80%.
- Some recent cogeneration plants have even higher utilization factors

An ideal cogeneration plant



A cogeneration plant with adjustable loads.

At times of high demand for process heat, all the steam is routed to the process-heating units and none to the condenser ($m_7 = 0$). The waste heat is zero in this mode.

If this is not sufficient, some steam leaving the boiler is throttled by an expansion or pressure reducing valve to the extraction pressure P_6 and is directed to the process-heating unit.

Maximum process heating is realized when all the steam leaving the boiler passes through the PRV ($m_5 = m_4$). No power is produced in this mode.

When there is no demand for process heat, all the steam passes through the turbine and the condenser ($m_5 = m_6 = 0$), and the cogeneration plant operates as an ordinary steam power plant.

$$\dot{Q}_{\text{in}} = \dot{m}_3(h_4 - h_3)$$

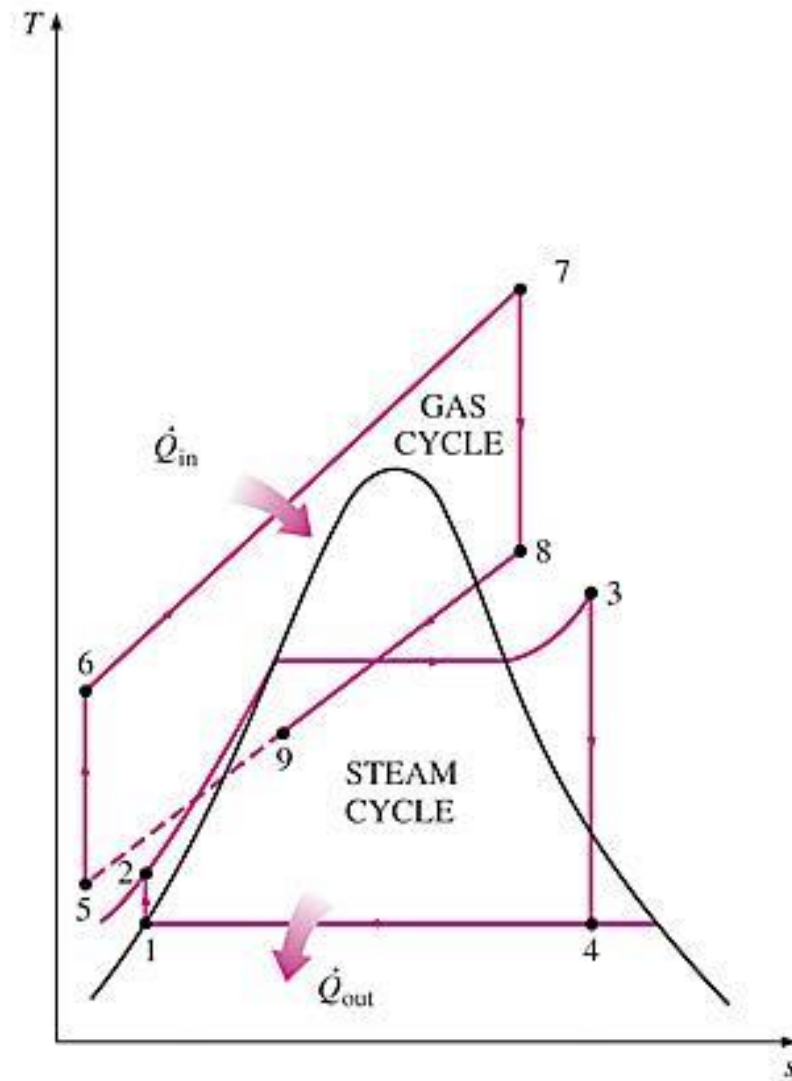
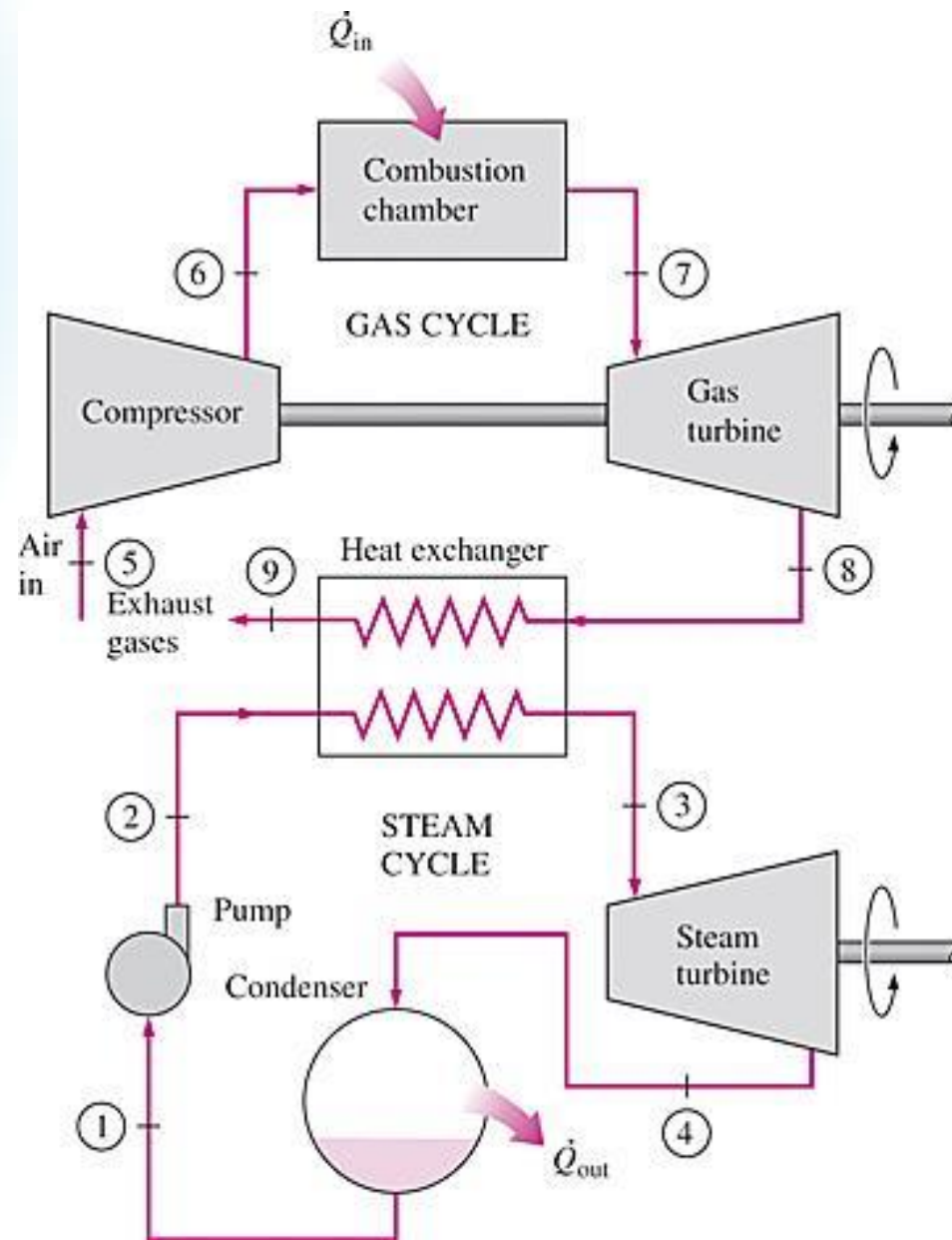
$$\dot{Q}_{\text{out}} = \dot{m}_7(h_7 - h_1)$$

$$\dot{Q}_p = \dot{m}_5 h_5 + \dot{m}_6 h_6 - \dot{m}_8 h_8$$

$$\dot{W}_{\text{turb}} = (\dot{m}_4 - \dot{m}_5)(h_4 - h_6) + \dot{m}_7(h_6 - h_7)$$

COMBINED GAS–VAPOR POWER CYCLES

- The continued quest for higher thermal efficiencies has resulted in rather innovative modifications to conventional power plants.
- A popular modification involves a gas power cycle topping a vapor power cycle, which is called the **combined gas–vapor cycle**, or just the **combined cycle**.
- The combined cycle of greatest interest is the gas-turbine (Brayton) cycle topping a steam-turbine (Rankine) cycle, which has a higher thermal efficiency than either of the cycles executed individually.
- It makes engineering sense to take advantage of the very desirable characteristics of the gas-turbine cycle at high temperatures *and* to use the high temperature exhaust gases as the energy source for the bottoming cycle such as a steam power cycle. The result is a combined gas–steam cycle.
- Recent developments in gas-turbine technology have made the combined gas–steam cycle economically very attractive.
- The combined cycle increases the efficiency without increasing the initial cost greatly. Consequently, many new power plants operate on combined cycles, and many more existing steam- or gas-turbine plants are being converted to combined-cycle power plants.
- Thermal efficiencies over 50% are reported



Combined gas-steam power plant.

PROBLEMS

1. Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 75 kPa. Determine the thermal efficiency of this cycle.
2. Consider a 210-MW steam power plant that operates on simple ideal Rankine cycle. Steam enters the turbine at 10 MPa and 500°C and is cooled in the condenser at a pressure of 10 kPa. Show the cycle on a T - s diagram with respect to saturation lines, and determine (a) the quality of the steam at the turbine exit, (b) the thermal efficiency of the cycle, and (c) the mass flow rate of the steam.
3. Repeat problem 2 assuming an isentropic efficiency of 85 percent for both the turbine and the pump.
4. Consider a steam power plant operating on the ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350°C and is condensed in the condenser at a pressure of 10 kPa. Determine (a) the thermal efficiency of this power plant, (b) the thermal efficiency if steam is superheated to 600°C instead of 350°C , and (c) the thermal efficiency if the boiler pressure is raised to 15 MPa while the turbine inlet temperature is maintained at 600°C .

5. Consider a steam power plant operating on the ideal Rankine cycle. Steam enters a high- pressure turbine at 15 Mpa and 600°C . and is condensed in the condenser pressure of 10 kpa. If the moisture content of the steam at the exit of the low- pressure turbine is not exceed 10.4 percent, determine (a) the pressure at which the steam should be reheated and (b) the thermal efficiency of the cycle. Assume the steam is reheated to inlet temperature of the high-pressure turbine.
6. Consider a steam power plant operating on the ideal regenerative Rankine cycle with one open feedwater heater. Steam enters the turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. Some steam leaves the turbine at a pressure of 1.2 MPa and enters the open feedwater heater. Determine the fraction of steam extracted from the turbine and the thermal efficiency of the cycle.

7. Consider a steam power plant that operates on an ideal reheat–regenerative Rankine cycle with one open feedwater heater, one closed feedwater heater, and one reheater. Steam enters the turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. Some steam is extracted from the turbine at 4 MPa for the closed feedwater heater, and the remaining steam is reheated at the same pressure to 600°C . The extracted steam is completely condensed in the heater and is pumped to 15 MPa before it mixes with the feedwater at the same pressure. Steam for the open feedwater heater is extracted from the low-pressure turbine at a pressure of 0.5 MPa. Determine the fractions of steam extracted from the turbine as well as the thermal efficiency of the cycle.